LINUX ASSIGNMENT

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Exercise 1 – Identify Your Linux Filesystem

Explore the storage configuration of your Linux system.

Determine:

1. The storage devices available.

The available storage devices can be identified using the lsblk command. The output shows the disks and other storage devices connected to the Linux system.

2. The partitions available on each device.

The lsblk command displays the partitions present on each storage device. The main disk is divided into separate partitions, which can be identified from the output.

3. The filesystem type used by each partition.

The filesystem type of each partition can be determined using lsblk -f. Common filesystem types include ext4 for Linux filesystems and swap for swap space.

4. The device/partition on which `/` is located.

The df -T command shows the device on which the root filesystem / is mounted. The partition listed beside / is the partition containing the root filesystem.

5. The filesystem type used for `/`.

The filesystem type for / can be identified using df -T. In a typical Kali Linux installation, the root filesystem uses ext4.

6. The total, used, and available storage space.

The df -h command displays the total filesystem capacity, used space, available space, and percentage of usage. The values should be recorded from the output of the user's own system.

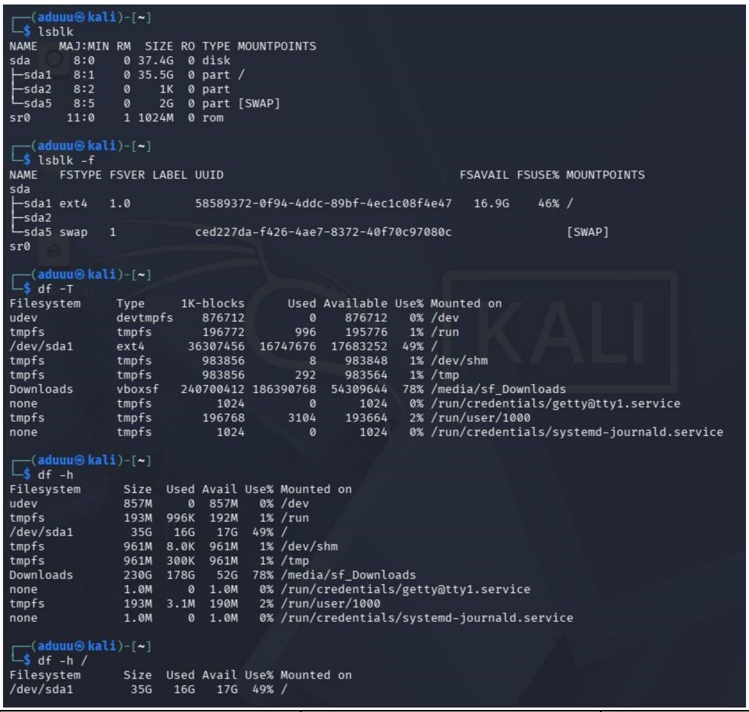
****Suggested commands:****

lsblk

lsblk -f

df -T df

-h



```
┌──(aduuu㉿kali)-[~]
└─$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 37.4G  0 disk
├─sda1   8:1    0 35.5G  0 part /
├─sda2   8:2    0    1K  0 part
└─sda5   8:5    0    2G  0 part [SWAP]
sr0     11:0    1 1024M  0 rom

┌──(aduuu㉿kali)-[~]
└─$ lsblk -f
NAME   FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1 ext4   1.0         58589372-0f94-4ddc-89bf-4ec1c08f4e47   16.9G    46% /
├─sda2
└─sda5 swap   1           ced227da-f426-4ae7-8372-40f70c97080c                [SWAP]
sr0

┌──(aduuu㉿kali)-[~]
└─$ df -T
Filesystem     Type      1K-blocks      Used Available Use% Mounted on
udev           devtmpfs     876712         0    876712   0% /dev
tmpfs          tmpfs        196772       996    195776   1% /run
/dev/sda1      ext4       36307456  16747676  17683252  49% /
tmpfs          tmpfs        983856         8    983848   1% /dev/shm
tmpfs          tmpfs        983856       292    983564   1% /tmp
Downloads      vboxsf    240700412 186390768  54309644  78% /media/sf_Downloads
none           tmpfs          1024         0      1024   0% /run/credentials/getty@tty1.service
tmpfs          tmpfs        196768      3104    193664   2% /run/user/1000
none           tmpfs          1024         0      1024   0% /run/credentials/systemd-journald.service

┌──(aduuu㉿kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            857M     0  857M   0% /dev
tmpfs           193M  996K  192M   1% /run
/dev/sda1        35G   16G   17G  49% /
tmpfs           961M  8.0K  961M   1% /dev/shm
tmpfs           961M  300K  961M   1% /tmp
Downloads       230G  178G   52G  78% /media/sf_Downloads
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           193M  3.1M  190M   2% /run/user/1000
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service

┌──(aduuu㉿kali)-[~]
└─$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        35G   16G   17G  49% /
```

Device/Partition	**Filesystem Type**	**Size**	**Mount Point**
/dev/sda1	Ext4	35.5G	/
/dev/sda2	No filesystem	1K	-
/dev/sda5	Swap	2G	[swap]
/dev/sda0	-	1024M	-

Exercise 2 – Disk → Partition → Filesystem → Mount Point

Using the information obtained in Exercise 1, draw the storage organization of your own Linux system.

```
sda (37.5G disk)
|
├── sda1 (35.5G partition)
|   ├── ext4 (filesystem)
|   └── / (mount point)
|
├── sda2 (1K partition)
|   └── No filesystem
|
└── sda5 (2G partition)
    ├── swap (filesystem)
```

└── [SWAP]

The physical disk sda is divided into separate partitions such as sda1, sda2, and sda5. The sda1 partition uses the ext4 filesystem and is mounted at /, while sda5 uses the swap filesystem for swap space. sda2 does not have a filesystem of its own.

Exercise 3 – Explore the Root Filesystem

Start from your current location.

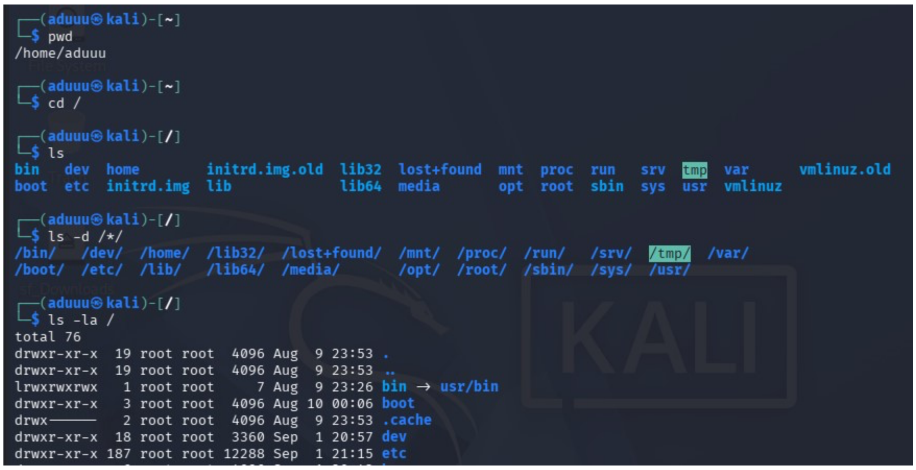
1. Determine your current working directory.

2. Navigate to the root directory.

3. Display the contents of the root directory.

4. Identify all directories directly available under `/`.

5. Determine whether hidden entries are present.



Exercise 4 – Linux Filesystem Hierarchy Investigation

For each location:

1. Examine its contents.

2. Identify at least three entries, where available.

3. Determine the likely purpose of the directory.

Directory	Example Contents	Purpose
/home	User directories, ., ..	Contains the personal home directories of regular users.
/root	.bashrc, .zshrc, .ssh	Contains the home directory and personal configuration files of the root (superuser) account.
/etc	passwd, shadow, hosts, fstab	Contains system-wide configuration files for the operating system and installed services.
/var	log, cache, spool	Contains variable data that changes during system operation, such as logs, caches and spool files.
/var/log	boot.log, wtmp, btmp, journal/	Contains system, boot, login and application logs used for troubleshooting and auditing.
/tmp	.X11-unix,ICE-unix, temporary directories	Provides temporary storage for temporary files and application/session data.
/usr	bin, lib, share	Contains installed programs, libraries and shared documentation/data.
/boot	Kernel files, initrd, grub/	Contains the kernel, initial RAM filesystem and bootloader files required to start the system.
/dev	null, zero, tty	Contains files representing hardware and virtual devices.
/proc	cpuinfo, meminfo, self, PID directories	A virtual, kernel-generated filesystem that provides live process and system information.

Exercise 5 – Find Your Home Directory

Without assuming its location:

1. Determine your username.

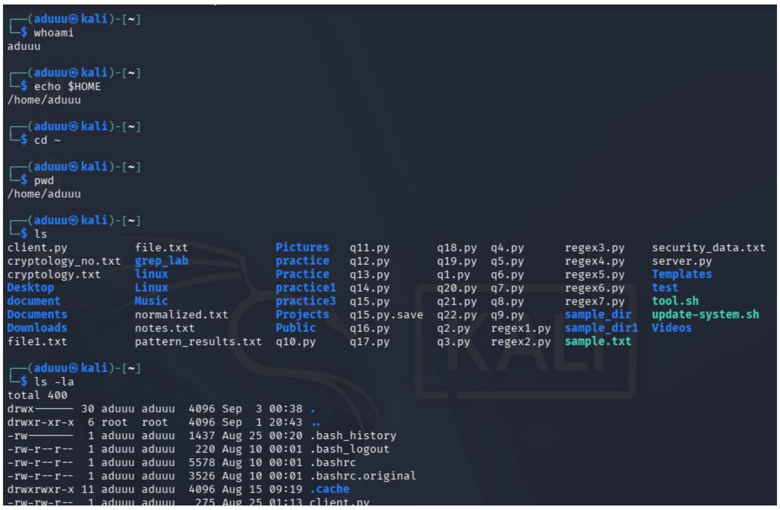
2. Determine your home directory.

3. Navigate to your home directory.

4. Display its absolute path.

5. List its normal contents.

6. List all contents, including hidden files.



```
┌──(aduuu㉿kali)-[~]
└─$ whoami
aduuu

┌──(aduuu㉿kali)-[~]
└─$ echo $HOME
/home/aduuu

┌──(aduuu㉿kali)-[~]
└─$ cd ~

┌──(aduuu㉿kali)-[~]
└─$ pwd
/home/aduuu

┌──(aduuu㉿kali)-[~]
└─$ ls
client.py         file.txt             Pictures   q11.py       q18.py  q4.py      regex3.py   security_data.txt
cryptology_no.txt grep_lab             practice   q12.py       q19.py  q5.py      regex4.py   server.py
cryptology.txt    linux                Practice   q13.py       q1.py   q6.py      regex5.py   Templates
Desktop           Linux                practice1  q14.py       q20.py  q7.py      regex6.py   test
document          Music                practice3  q15.py       q21.py  q8.py      regex7.py   tool.sh
Documents         normalized.txt       Projects   q15.py.save  q22.py  q9.py      sample_dir  update-system.sh
Downloads         notes.txt            Public     q16.py       q2.py   regex1.py  sample_dir1 Videos
file1.txt         pattern_results.txt  q10.py     q17.py       q3.py   regex2.py  sample.txt

┌──(aduuu㉿kali)-[~]
└─$ ls -la
total 400
drwx------ 30 aduuu aduuu 4096 Sep  3 00:38 .
drwxr-xr-x  6 root  root  4096 Sep  1 20:43 ..
-rw-------  1 aduuu aduuu 1437 Aug 25 00:20 .bash_history
-rw-r--r--  1 aduuu aduuu  220 Aug 10 00:01 .bash_logout
-rw-r--r--  1 aduuu aduuu 5578 Aug 10 00:01 .bashrc
-rw-r--r--  1 aduuu aduuu 3526 Aug 10 00:01 .bashrc.original
drwxrwxr-x 11 aduuu aduuu 4096 Aug 15 09:19 .cache
-rw-rw-r--  1 aduuu aduuu  275 Aug 25 01:13 client.py
```

Exercise 6 – Hidden Files Investigation

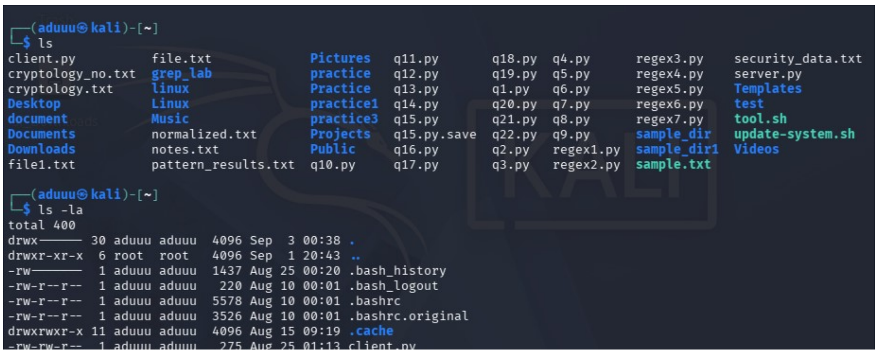
Examine your home directory.

Compare:

ls ls

-a

Identify at least 2 hidden files/directories.



```
┌──(aduuu㉿kali)-[~]
└─$ ls
client.py         file.txt             Pictures   q11.py       q18.py  q4.py      regex3.py   security_data.txt
cryptology_no.txt grep_lab             practice   q12.py       q19.py  q5.py      regex4.py   server.py
cryptology.txt    linux                Practice   q13.py       q1.py   q6.py      regex5.py   Templates
Desktop           Linux                practice1  q14.py       q20.py  q7.py      regex6.py   test
document          Music                practice3  q15.py       q21.py  q8.py      regex7.py   tool.sh
Documents         normalized.txt       Projects   q15.py.save  q22.py  q9.py      sample_dir  update-system.sh
Downloads         notes.txt            Public     q16.py       q2.py   regex1.py  sample_dir1 Videos
file1.txt         pattern_results.txt  q10.py     q17.py       q3.py   regex2.py  sample.txt

┌──(aduuu㉿kali)-[~]
└─$ ls -la
total 400
drwx------ 30 aduuu aduuu 4096 Sep  3 00:38 .
drwxr-xr-x  6 root  root  4096 Sep  1 20:43 ..
-rw-------  1 aduuu aduuu 1437 Aug 25 00:20 .bash_history
-rw-r--r--  1 aduuu aduuu  220 Aug 10 00:01 .bash_logout
-rw-r--r--  1 aduuu aduuu 5578 Aug 10 00:01 .bashrc
-rw-r--r--  1 aduuu aduuu 3526 Aug 10 00:01 .bashrc.original
drwxrwxr-x 11 aduuu aduuu 4096 Aug 15 09:19 .cache
-rw-rw-r--  1 aduuu aduuu  275 Aug 25 01:13 client.py
```

Answer:

1. How does Linux identify a hidden file?

Any file or directory whose name starts with a dot (.) is treated as hidden in Linux.

2. Why does a normal `ls` not display these files?

The normal ls command does not show files that start with a dot by default. This is simply to keep the usual listing clean and avoid showing configuration and other hidden files. It is only a display convention, not a security feature.

3. What do `.` and `..` mean?

. represents the current directory, while .. represents the parent directory.

4. Can a hidden file still be accessed if its name is known?

Yes. A hidden file can still be accessed normally if you know its name. The dot at the beginning only makes it hidden from the usual ls listing; it does not stop you from opening or using the file.

5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?

Hidden files are not always safe or harmful just because they are hidden. Some, like .bashrc and .bash_history, are normal system files, while .ssh can contain sensitive information. So, an investigator should check the file's contents, permissions, and purpose before deciding whether it is suspicious.

Exercise 7 – Build a Small Filesystem Hierarchy

Explore: mkdir

Hint: you can create this hierarchy in a single command

Create the following files:

cyber_lab/evidence/network/traffic.txt

cyber_lab/evidence/system/processes.txt

cyber_lab/logs/security.log cyber_lab/scripts/check.sh

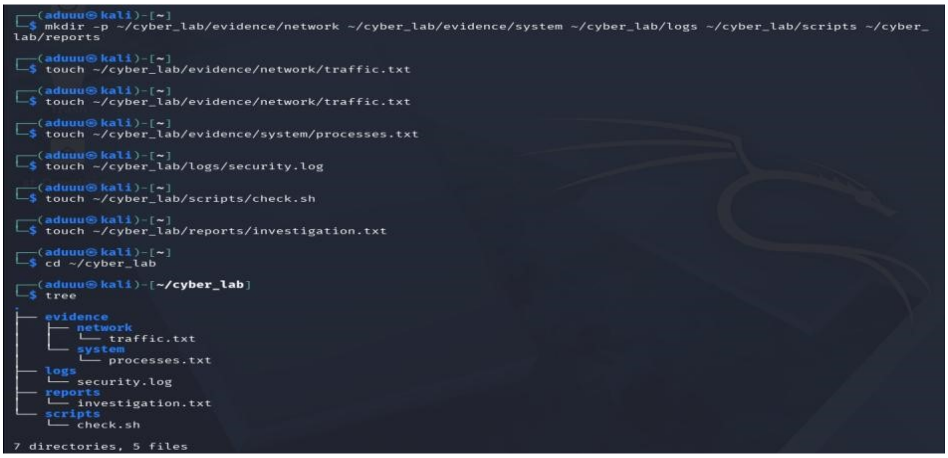
cyber_lab/reports/investigation.txt

Use appropriate Linux commands to create the structure.

Verify that the required hierarchy has been created successfully.

Hint: tree

Explore options



Exercise 8 – Absolute and Relative Paths

Assume your current directory is:

~/cyber_lab

Perform the following:

1. Navigate to `evidence/network` using a relative path.

2. Navigate to `/etc` using an absolute path.

3. Return to your home directory using the shortest possible command.

4. Navigate to `cyber_lab/reports`.

5. Move one directory upward.

6. Display the current directory after every operation.

*** What is an absolute path?**

An absolute path starts from the root directory / and gives the complete location of a file or directory. For example: /home/aduuu/cyber_lab/evidence/network

*** What is a relative path?**

A relative path specifies a location based on the current working directory. For example:

evidence/network.

*** What is the difference between `/` and `~`?**

/ : root directory of the entire Linux filesystem.

~ : shortcut for the current user's home directory.

Exercise 9 – Identify File Types

Investigate the following:

/etc/passwd

/etc

/bin/bash

/dev/null

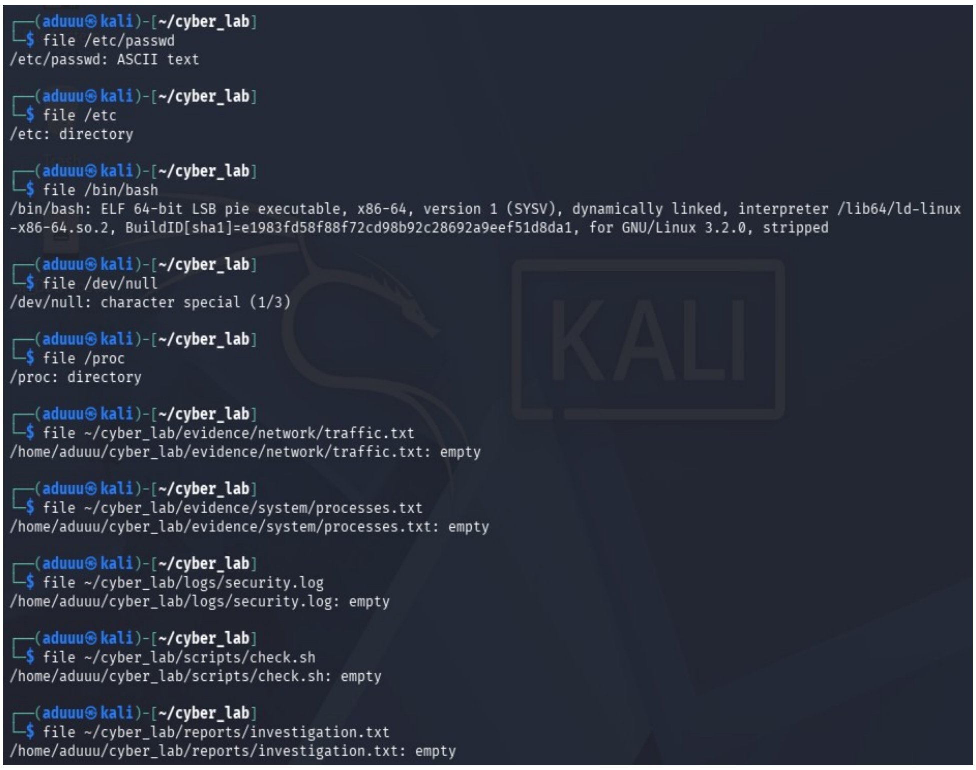
/proc

Also investigate the files created in Exercise 7.

file <filename> ls

-l <filename>

Classify each object.



```
┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file /etc/passwd
/etc/passwd: ASCII text

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file /etc
/etc: directory

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file /bin/bash
/bin/bash: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=e1983fd58f88f72cd98b92c28692a9eef51d8da1, for GNU/Linux 3.2.0, stripped

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file /dev/null
/dev/null: character special (1/3)

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file /proc
/proc: directory

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file ~/cyber_lab/evidence/network/traffic.txt
/home/aduuu/cyber_lab/evidence/network/traffic.txt: empty

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file ~/cyber_lab/evidence/system/processes.txt
/home/aduuu/cyber_lab/evidence/system/processes.txt: empty

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file ~/cyber_lab/logs/security.log
/home/aduuu/cyber_lab/logs/security.log: empty

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file ~/cyber_lab/scripts/check.sh
/home/aduuu/cyber_lab/scripts/check.sh: empty

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file ~/cyber_lab/reports/investigation.txt
/home/aduuu/cyber_lab/reports/investigation.txt: empty
```

Object	**Type**
/etc/passwd	Regular text file
/etc	Directory
/bin/bash	Executable files
/dev/null	Character special device file
/proc	Virtual/pseudo filesystem directory
traffic.txt	Regular files
processes.txt	Regular files
Security.log	Regular files
check.sh	Regular files
investigation.txt	Regular files

Exercise 10 – Examine File Metadata

Use:~/cyber_lab/reports/investigation.txt

Add a small amount of text to the file.

Examine it using:

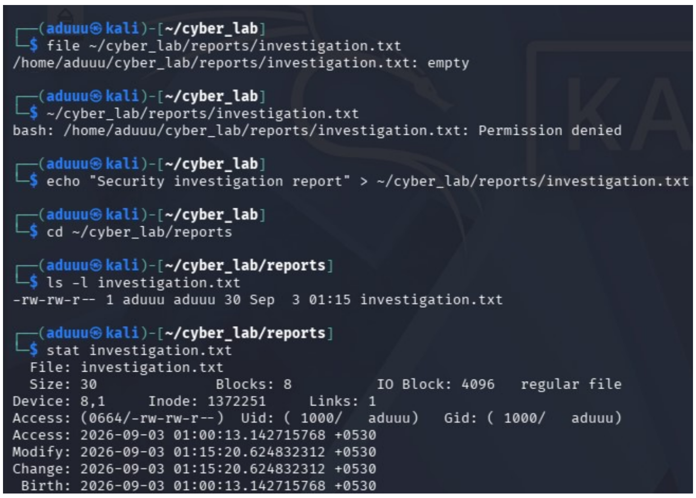
ls -l

stat investigation.txt

Identify:

* **Filename**
* **File size**
* **File type**
* **Owner**
* **Group**
* **Inode number**
* **Access time**
* **Modification time**
* **Change time**

Answer:



```
┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ file ~/cyber_lab/reports/investigation.txt
/home/aduuu/cyber_lab/reports/investigation.txt: empty

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ ~/cyber_lab/reports/investigation.txt
bash: /home/aduuu/cyber_lab/reports/investigation.txt: Permission denied

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ echo "Security investigation report" > ~/cyber_lab/reports/investigation.txt

┌──(aduuu㉿kali)-[~/cyber_lab]
└─$ cd ~/cyber_lab/reports

┌──(aduuu㉿kali)-[~/cyber_lab/reports]
└─$ ls -l investigation.txt
-rw-rw-r-- 1 aduuu aduuu 30 Sep  3 01:15 investigation.txt

┌──(aduuu㉿kali)-[~/cyber_lab/reports]
└─$ stat investigation.txt
  File: investigation.txt
  Size: 30              Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 1372251     Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   aduuu)   Gid: ( 1000/   aduuu)
Access: 2026-09-03 01:00:13.142715768 +0530
Modify: 2026-09-03 01:15:20.624832312 +0530
Change: 2026-09-03 01:15:20.624832312 +0530
 Birth: 2026-09-03 01:00:13.142715768 +0530
```

1. What is metadata?

Metadata is information about a file, such as its size, owner, permissions, inode number, and timestamps.

2. Is metadata the same as file content?

No. Metadata describes the file, while file content is the actual data stored inside the file.

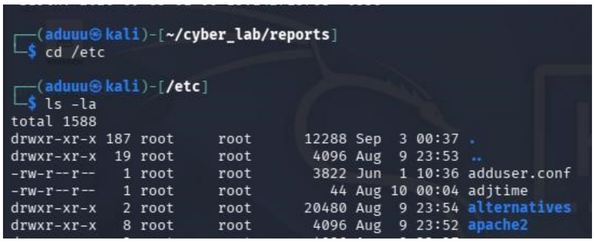
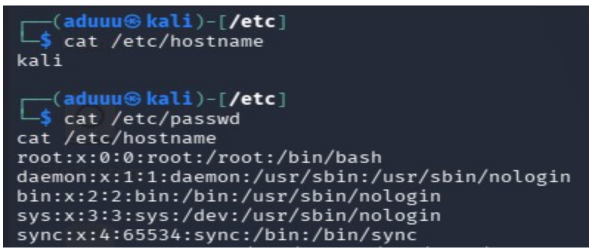
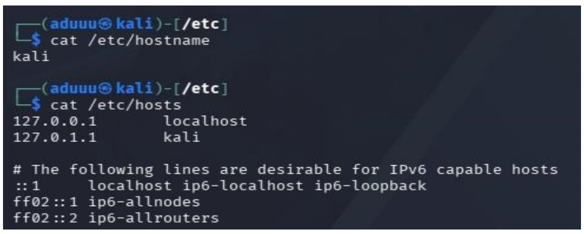
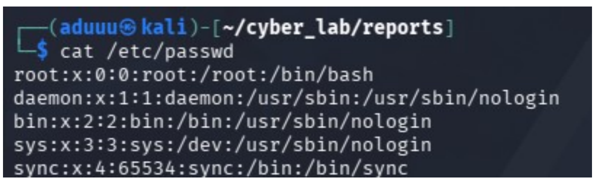
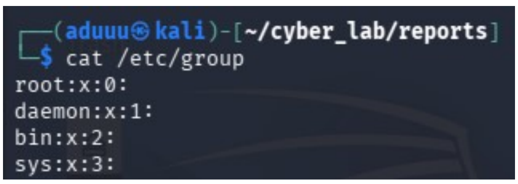
3. Which timestamp changes when file contents are modified?

The modification time (mtime) changes when the contents of a file are modified.

Exercise 11 – Explore `/etc`

Investigate:

/etc

```

**Locate:**

**passwd group**

**hostname**

**hosts**

**Note: Without modifying any system files, examine their contents where permitted.**

**Answer:**
















**1. What kind of information is primarily stored under `/etc`?**
```

/etc mainly stores system settings and configuration files.

2. What information does `/etc/passwd` contain?

/etc/passwd contains information about the users on the system.

3. What information does `/etc/hostname` contain?

/etc/hostname contains the name of the computer/system.

4. Why is `/etc` security-sensitive?

/etc is security-sensitive because it contains important system and user settings.

Exercise 12 – Explore Linux Log Storage

Investigate:

/var/log

\`\`\`

Identify:

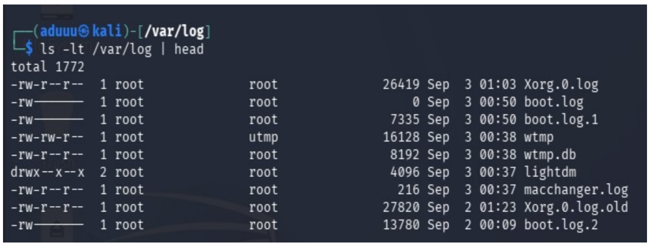
*** Available log files**

*** Available log directories**

*** Recently modified entries**

*** Entries you can read**

*** Entries you cannot read as a normal user Answer:**



```
┌──(aduuu㉿kali)-[/var/log]
└─$ ls -lt /var/log | head
total 1772
-rw-r--r--  1 root    root     26419 Sep  3 01:03 Xorg.0.log
-rw-------  1 root    root         0 Sep  3 00:50 boot.log
-rw-------  1 root    root      7335 Sep  3 00:50 boot.log.1
-rw-rw-r--  1 root    utmp     16128 Sep  3 00:38 wtmp
-rw-r--r--  1 root    root      8192 Sep  3 00:38 wtmp.db
drwx--x--x  2 root    root      4096 Sep  3 00:37 lightdm
-rw-r--r--  1 root    root       216 Sep  3 00:37 macchanger.log
-rw-r--r--  1 root    root     27820 Sep  2 01:23 Xorg.0.log.old
-rw-------  1 root    root     13780 Sep  2 00:09 boot.log.2
```

Identify

- Available log files: Files like syslog, auth.log, kern.log, and dpkg.log may be present depending on the system.
- Available log directories: Folders such as apache2 and journal may be present.
- Recently modified entries: Use ls -lt to see the entries that were modified most recently at the top.
- Entries you can read: Log files that your user has permission to read.

- Entries you cannot read as a normal user: Log files that your user does not have permission to read, usually because they contain sensitive system or security information.

1. Why does Linux maintain logs?

Linux maintains logs to record system and user activities, errors, and events.

2. Why are logs useful during incident investigation?

Logs help investigators find out what happened and when it happened.

3. Why might access to some logs be restricted?

Some logs may contain sensitive user or security information, so access is restricted to protect them.

Exercise 13 – Investigate `/tmp`

Explore `/tmp` without modifying files belonging to other users.

Determine:

1. Whether files/directories currently exist.

Yes. /tmp contains several files and directories used for temporary files, sockets, and application/session data.

2. Whether hidden files exist.

Yes. Hidden entries such as .X11-unix, .ICE-unix, and other dot-prefixed entries may be present.

3. Who owns some of the entries.

The entries may be owned by root or by the current user, depending on which process created them.

4. Which entries were modified recently.

The command ls -lt /tmp shows the entries in order of modification time, with the most recently modified entries appearing first.

```
┌──(aduuu㉿kali)-[~]
└─$ cd /tmp

┌──(aduuu㉿kali)-[/tmp]
└─$ ls -la
total 12
drwxrwxrwt 11 root  root   300 Sep  3 01:47 .
drwxr-xr-x 19 root  root  4096 Aug  9 23:53 ..
-rw-------  1 aduuu aduuu    0 Sep  3 00:38 config-err-POoUuI
drwxrwxrwt  2 root  root    40 Sep  3 00:37 .font-unix
drwxrwxrwt  2 root  root    60 Sep  3 00:38 .ICE-unix
srw-rw-rw-  1 root  root     0 Sep  3 00:37 .iprt-localipc-DRMIpcServer
drwx------  3 root  root    60 Sep  3 00:38 systemd-private-363ea4e92e194e049acfbb042e05ec9b-colord.service-FWUzJP
drwx------  3 root  root    60 Sep  3 00:37 systemd-private-363ea4e92e194e049acfbb042e05ec9b-ModemManager.service-3
vRgRJ
drwx------  3 root  root    60 Sep  3 00:37 systemd-private-363ea4e92e194e049acfbb042e05ec9b-polkit.service-Euknvo
drwx------  3 root  root    60 Sep  3 00:37 systemd-private-363ea4e92e194e049acfbb042e05ec9b-systemd-logind.service
-X2i8Hy
drwx------  3 root  root    60 Sep  3 00:38 systemd-private-363ea4e92e194e049acfbb042e05ec9b-upower.service-AbeEKK
-r--r--r--  1 root  root    11 Sep  3 00:37 .X0-lock
drwxrwxrwt  2 root  root    60 Sep  3 00:37 .X11-unix
-rw-------  1 aduuu aduuu  398 Sep  3 00:38 .xfsm-ICE-CQLAV3
drwxrwxrwt  2 root  root    40 Sep  3 00:37 .XIM-unix

┌──(aduuu㉿kali)-[/tmp]
└─$ ls -lt /tmp
total 0
drwx------ 3 root  root  60 Sep  3 00:38 systemd-private-363ea4e92e194e049acfbb042e05ec9b-colord.service-FWUzJP
drwx------ 3 root  root  60 Sep  3 00:38 systemd-private-363ea4e92e194e049acfbb042e05ec9b-upower.service-AbeEKK
-rw------- 1 aduuu aduuu  0 Sep  3 00:38 config-err-POoUuI
drwx------ 3 root  root  60 Sep  3 00:37 systemd-private-363ea4e92e194e049acfbb042e05ec9b-ModemManager.service-3vRg
RJ
drwx------ 3 root  root  60 Sep  3 00:37 systemd-private-363ea4e92e194e049acfbb042e05ec9b-systemd-logind.service-X2
i8Hy
drwx------ 3 root  root  60 Sep  3 00:37 systemd-private-363ea4e92e194e049acfbb042e05ec9b-polkit.service-Euknvo
```

Answer:

Why can temporary directories be relevant during a cybersecurity investigation?

/tmp is important because programs commonly use it to create temporary files. During a security investigation, suspicious scripts, downloaded files, extracted archives, or malwarerelated files may be found there, so it should be examined carefully.

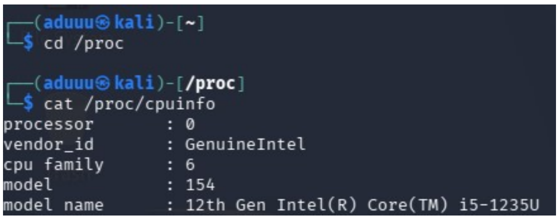
Exercise 14 – Explore the `/proc` Filesystem

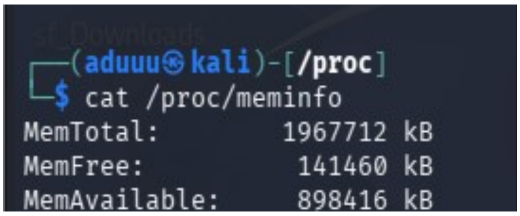
Investigate:

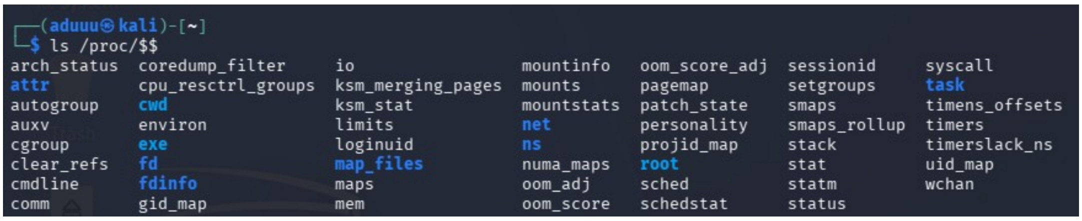
/proc

```

Examine:

cat /proc/cpuinfo cat

/proc/meminfo ```

Then identify your current shell PID: echo

$$

```

Check whether a corresponding directory exists under `/proc`.







Answer:

1. What information does `/proc/cpuinfo` provide?

It provides information about the CPU, such as its model, speed, cache size, and supported features.

2. What information does `/proc/meminfo` provide?

It provides information about the system's current memory, including RAM and swap memory.

3. Why do directories with numbers appear inside `/proc`?

The numbered directories represent process IDs (PIDs) of currently running processes.

4. Is `/proc` an ordinary disk directory?

No. /proc is created by the Linux kernel and provides information about the system and running processes.

5. Why could `/proc` be useful during security investigation?

It shows information about currently running processes and system activity, which can help investigators identify suspicious processes or activity.

Exercise 15 – Explore `/dev`

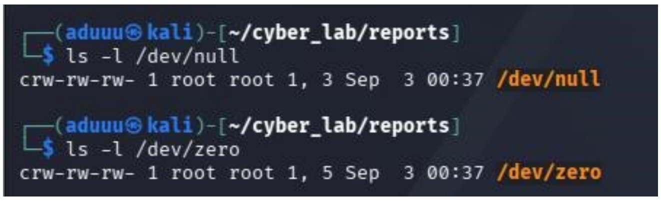
/dev

```

**Examine:**
```

ls -l /dev/null ls

-l /dev/zero

` ` `



Answer:

1. What does `/dev` represent?

It is where Linux provides hardware and virtual devices as files. Programs can interact with devices such as disks and terminals through these files.

2. Are the entries ordinary user files?

No. They are character devices, not regular files. The c at the beginning of the permissions shows that they are character devices.

3. Determine the file type of `/dev/null`.

/dev/null is a character special device with major number 1 and minor number 3.

4. Why does Linux expose devices through the filesystem?

Linux treats devices much like files, which makes it easy for programs to interact with them. For example, /dev/null simply throws away anything sent to it, while /dev/zero keeps giving zeros whenever it is read.

Exercise 16 – Filesystem Space Investigation

Determine:

1. Total filesystem capacity

35G

2. Used space.

16G

3. Available space.

17G

4. Percentage of filesystem utilization.

49%

5. Space occupied by your home directory.

152M

6. Space occupied by the `cyber_lab` directory.

32K

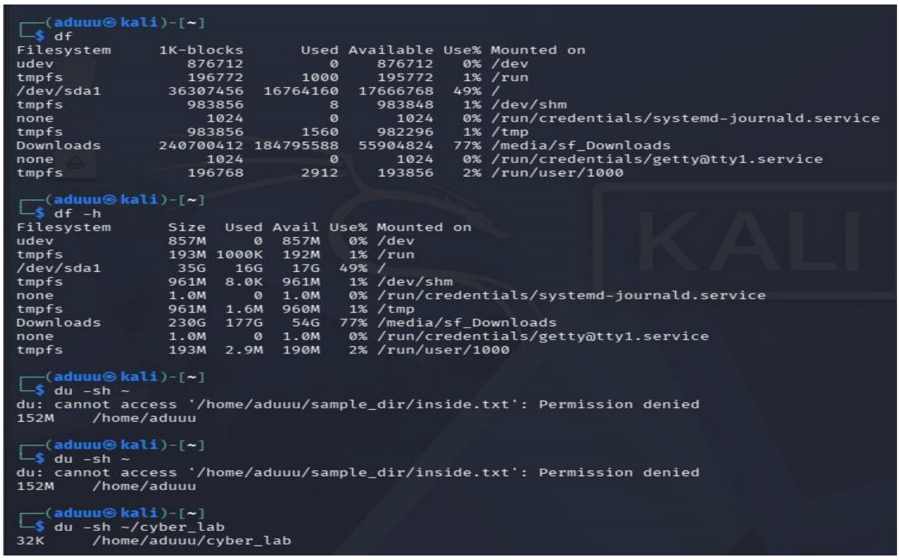
Use appropriate combinations of:

df df -

h du

du -h

du -sh



Answer:

Explain the difference between `df` and `du`.

- df tells you how full the whole filesystem is. In your case, /dev/sda1 is 49% full, with 17G available.
- du tells you how much space files and folders are using. Your home directory uses 152M, while cyber_lab uses only 32K.

Exercise 17 – Cybersecurity Investigation Scenario

Assume that you are investigating a Linux machine after suspicious activity has been reported.

You are initially asked to examine only the filesystem.

Identify which of the following locations you would investigate and explain **why:**

/home

/etc

/var/log

/tmp

/proc

/dev

/root

Answers:

/home : To look for suspicious files, scripts, downloads, hidden files, and user activity.

/etc : To check for changed system settings, new users, sudo changes, or scheduled tasks.

/var/log : To find login activity, errors, and other clues about what happened.

/tmp : Attackers may use it to temporarily store malicious files or scripts.

/proc : It shows what is currently running on the system, which can help identify suspicious processes.

/root : If accessible, it may show signs that an attacker gained root/administrator access.

/dev : low priority : /dev normally contains system device files and is less likely to contain useful evidence.